Binary LP Relaxation as Min-Cut

Overview

Polynomial solvability

Min-cut

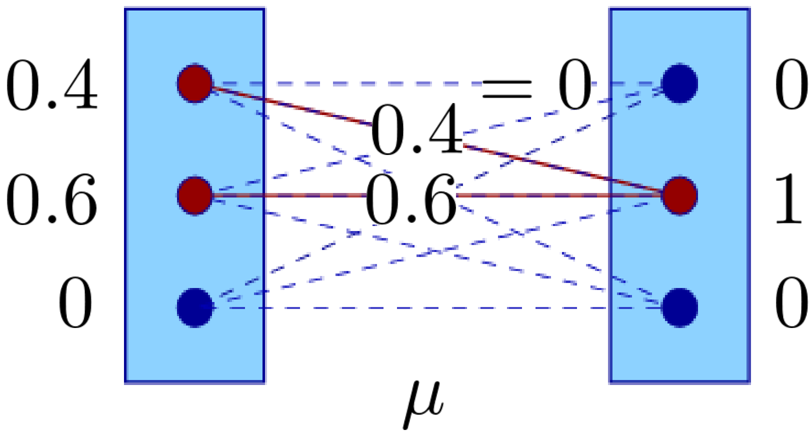
Submodularity

Non-negative edge costs

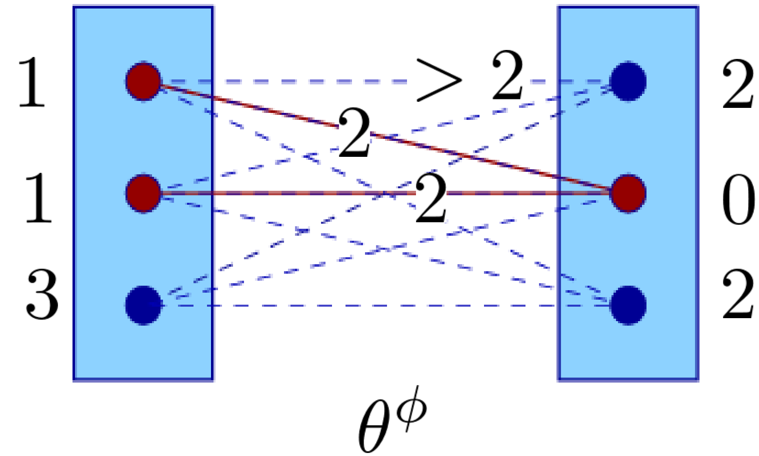
Binary energy

Recall: Dual Optimaity

Primal:



Dual:

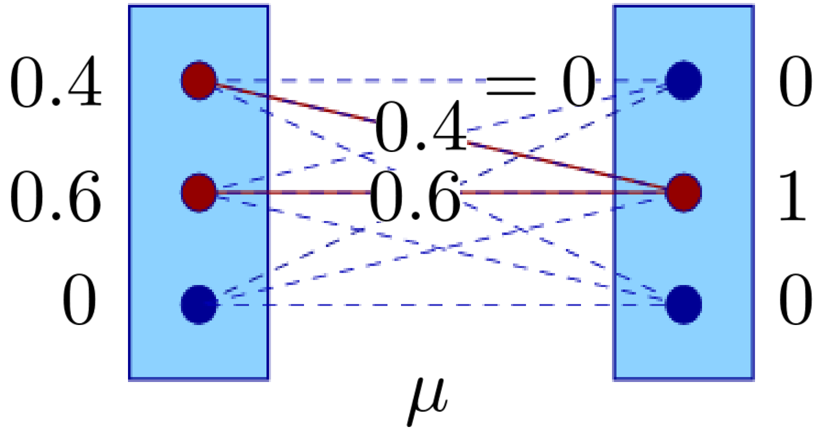


Non-zero weights of primal variables can be assigned to:

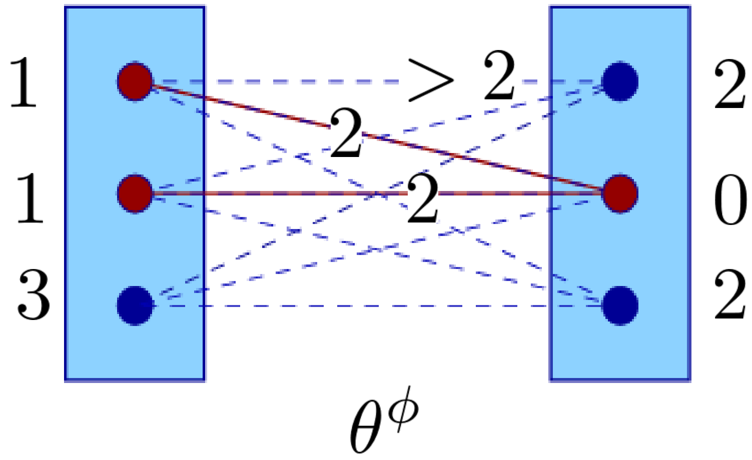
1) only locally minimal labels and label pairs
2) any except locally minimal labels and label pairs
3) must be assigned to all locally minimal labels and label pairs
4) no idea

Recall: Dual Optimaity

Primal:



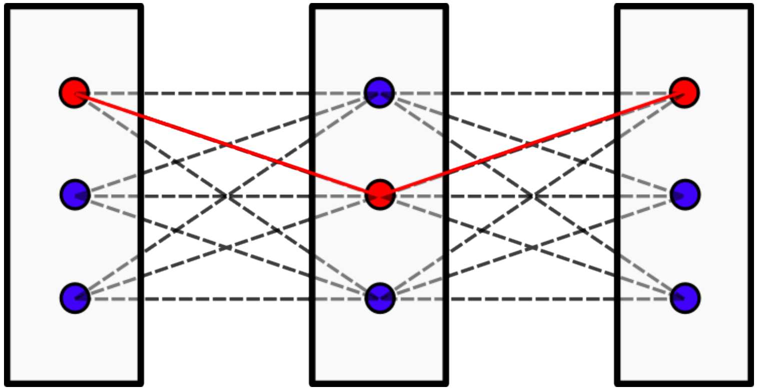
Dual:



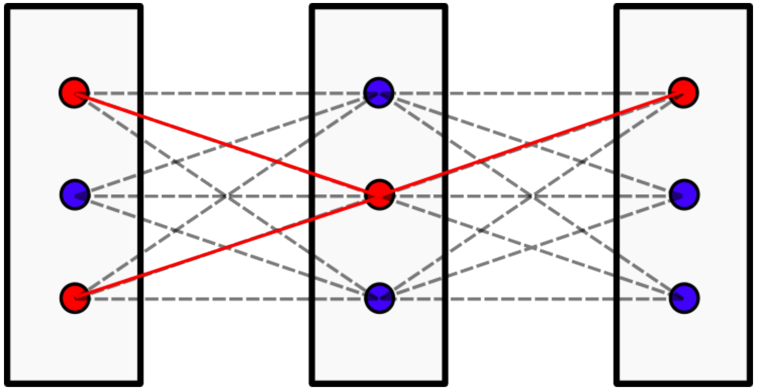
Existence of a relaxed solution consistent with the reparametrization means:

1) optimality of the reparametrization
2) nothing
3) existence of an optimal integer labeling consistent with the reparametrization
4) no idea

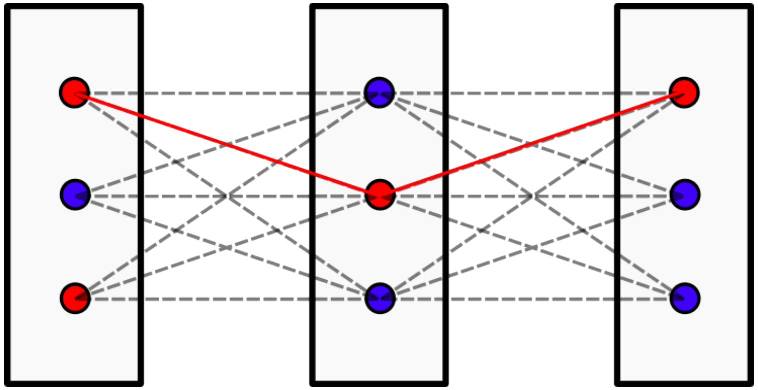
Recall: Arc Consistency



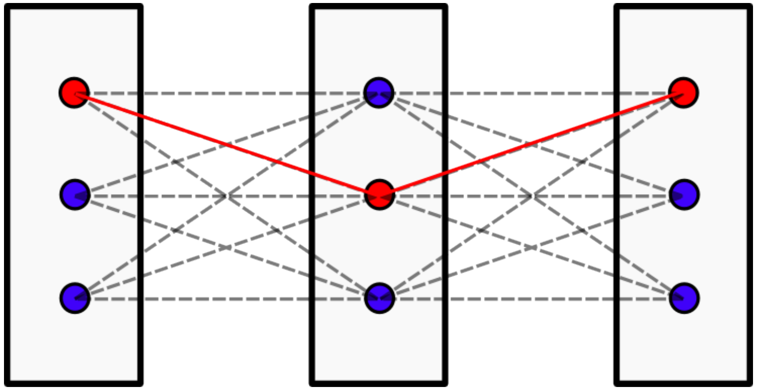
strict arc-consistency



arc-consistency



no arc-consistency



kernel

Recall: Arc Consistency

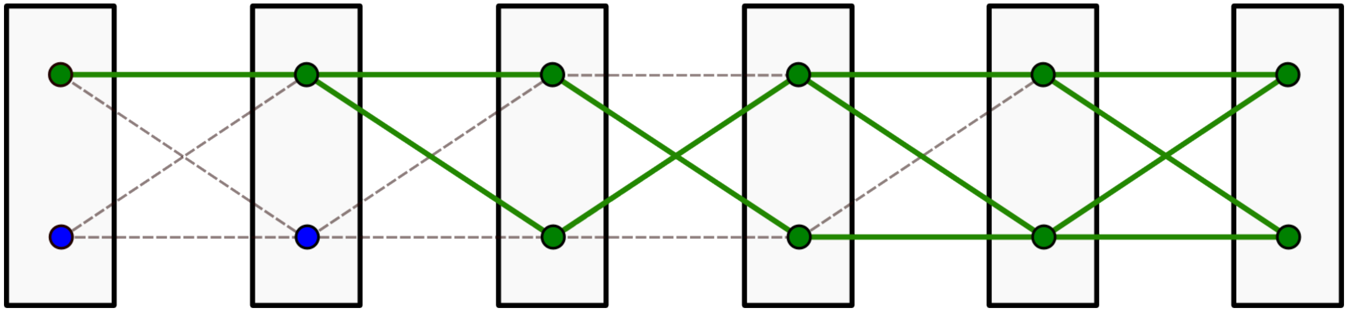
Strict arc-consistency implies:

1) dual optimality (optimality of the reparametrization)
2) an optimal integer labeling can be easily reconstructed
3) none of 1)-2)
4) both 1)+2)
5) no idea

Recall: Arc Consistency

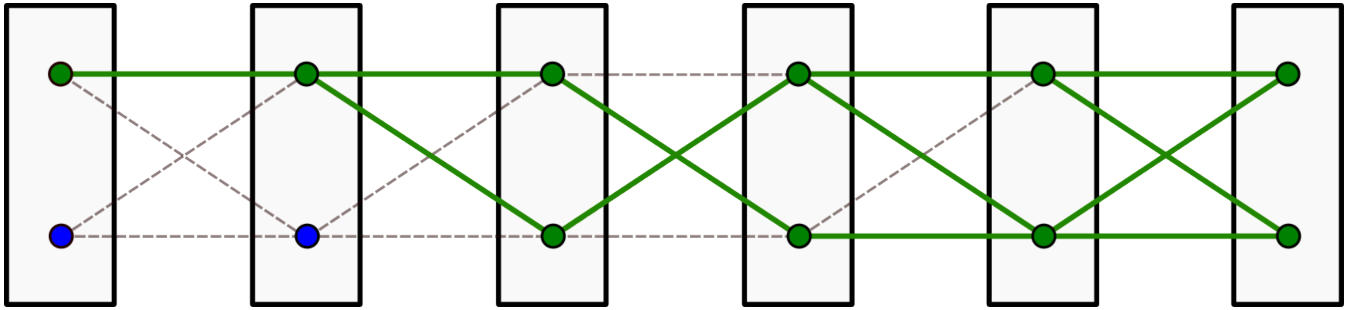
~~Strict~~ arc-consistency implies:

1) dual optimality (optimality of the reparametrization)
2) an optimal integer labeling can be easily reconstructed
3) a consistent optimal relaxed labeling can be reconstructed
4) none of 1)-3)
5) no idea



Theorem. Let θ^ϕ have a non-empty kernel. Then there is a primal solution with coordinates from $\{0, \frac{1}{2}, 1\}^I$ consistent with the locally minimal labels and label pairs.

Arc Consistency => Optimality and Half-Integrality



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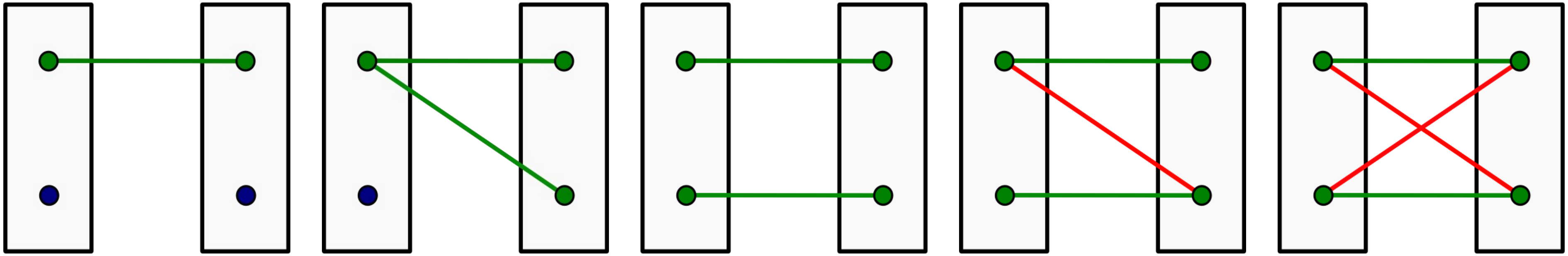

Basis Primal Solutions for Pairwise Factors

Unary basis solutions: $\mu \in \{(0,1),(1,0),(\frac{1}{2},\frac{1}{2})\}$

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Pairwise: **?**

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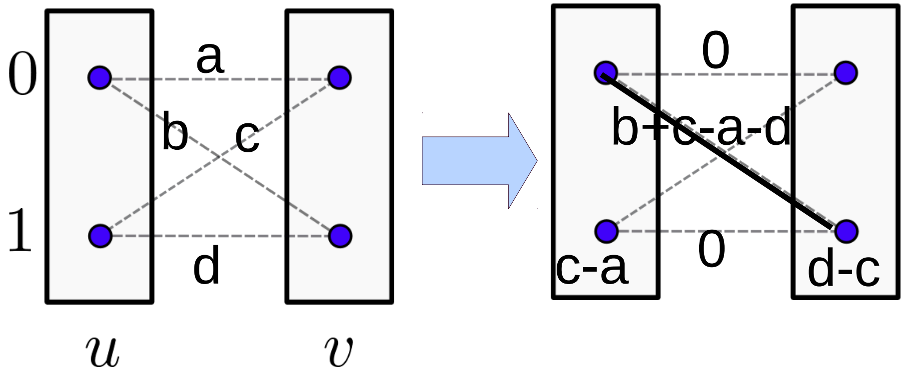
Pairwise: ?

Recall:

Diagonal form:



$c_{uv}\bar{x}_u x_v$

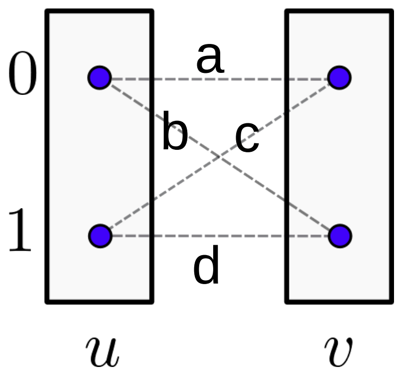
submodular: $c_{uv} \geq 0$

Basis Primal Solutions for Pairwise Factors

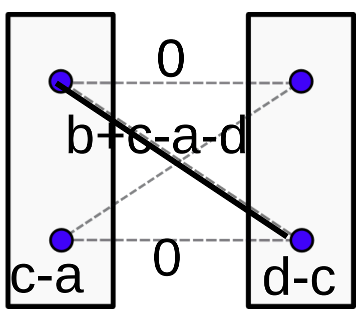
Unary basis solutions: $\mu \in \{(0,1),(1,0),(\frac{1}{2},\frac{1}{2})\}$

Pairwise: **?**

Recall:



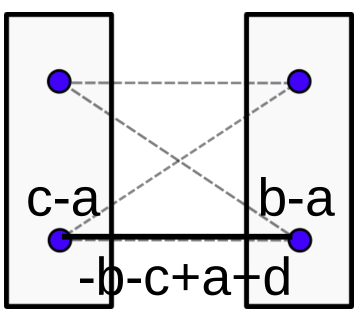
Diagonal form:



$c_{uv}\bar{x}_u x_v$

Or
(similarly)

Parallel form:



$c_{uv}x_u x_v$

$$\text{submodular: } c_{uv} \geq 0 \qquad \text{supermodular: } c_{uv} \geq 0$$

Basis Primal Solutions for Pairwise Factors

Unary basis solutions: $\mu \in \{(0,1),(1,0),(\frac{1}{2},\frac{1}{2})\}$

Pairwise: given μ_v, the optimization w.r.t. μ_{uv} is independent for each uv :

$$\min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle = \min_{\mu_v, v \in \mathcal{V}} \sum_{uv \in \mathcal{E}} \min_{\mu_{uv} \in \mathcal{L}_{uv}(\mu_u, \mu_v)} \langle \theta_{uv}, \mu_{uv} \rangle$$

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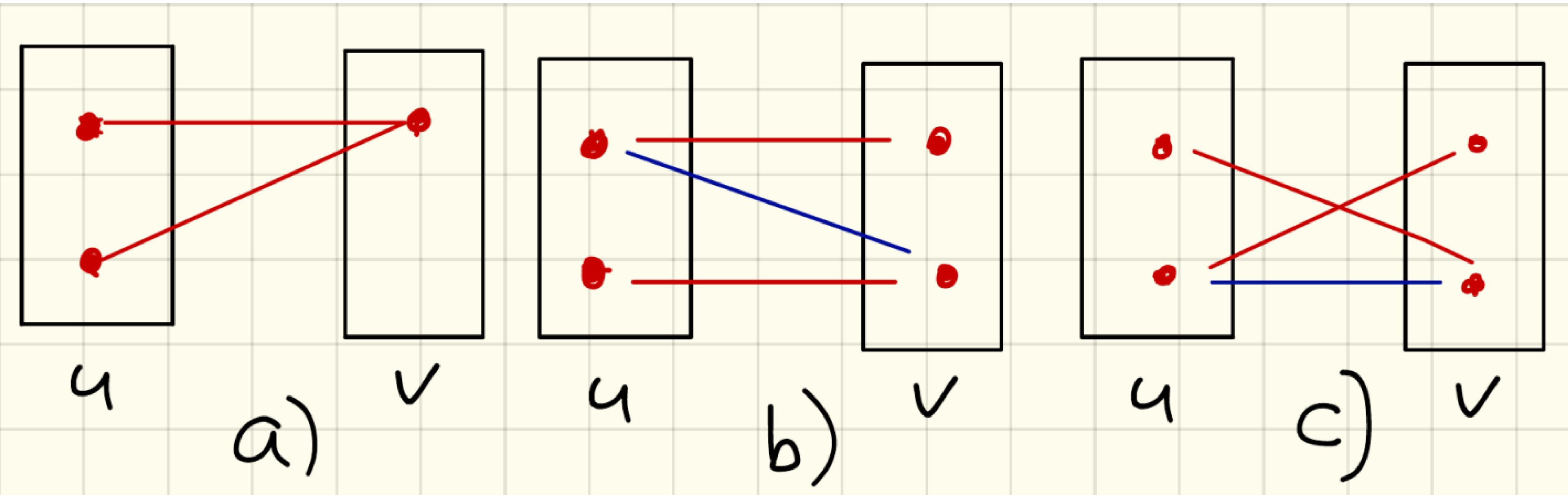
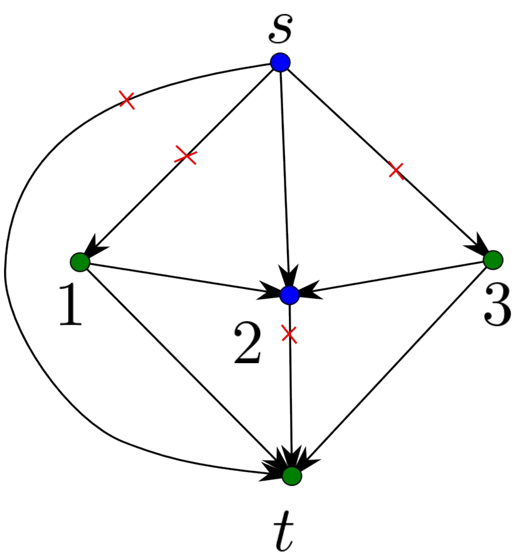
Optimal primal configurations:



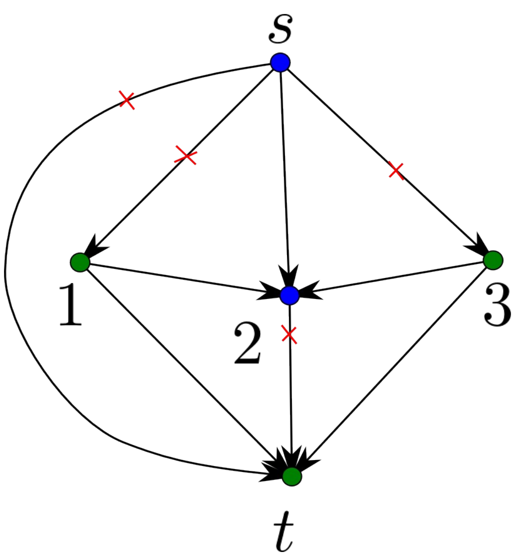
Transformation of Posiforms: QPBF → O-QPBF

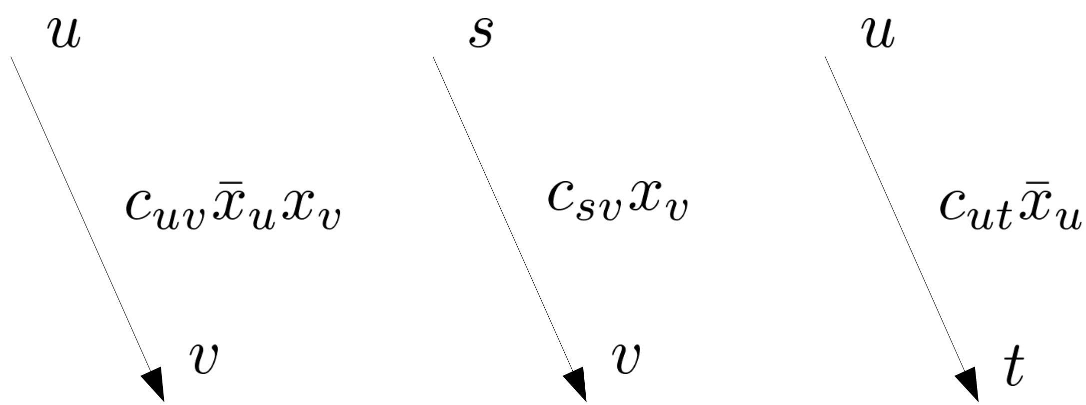


$$u \xrightarrow{c_{uv}\bar{x}_u x_v} v \qquad s \xrightarrow{c_{sv} x_v} v \qquad u \xrightarrow{c_{ut}\bar{x}_u} t$$

Solvable: $c_{uv} \geq 0$

Transformation of Posiforms: QPBF → O-QPBF

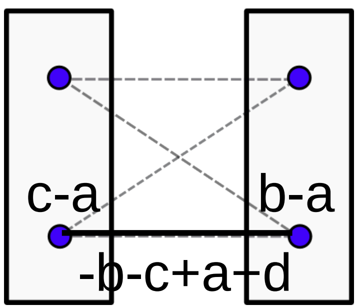




Solvable: $c_{uv} \geq 0$

Problem:

Parallel form:



$c_{uv}x_u x_v$

supermodular: $c_{uv} \geq 0$

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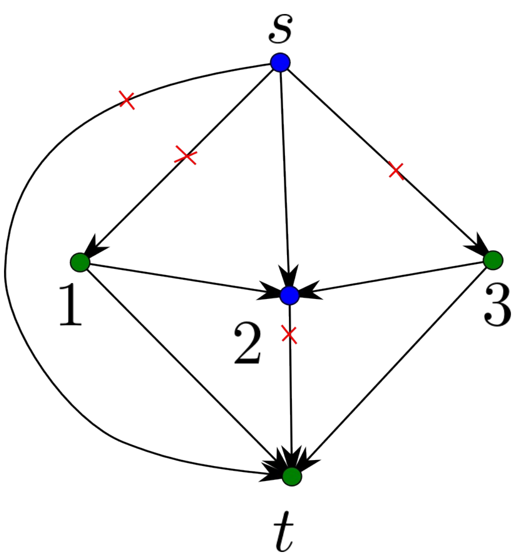

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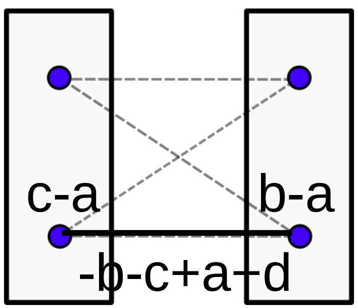

$c_{uv} x_u x_v$

supermodular: $c_{uv} \geq 0$

Idea:

$$c_{uv} x_u x_v \to \frac{1}{2} c_{uv} x_u \bar{y}_v + \frac{1}{2} c_{uv} \bar{y}_u x_v$$

$$\text{s.t. } x_u = \bar{y}_u, \;\; u \in \mathcal{V}$$

Transformation of Posiforms: QPBF → O-QPBF

Unary terms:

$$c_v x_v \to \frac{1}{2} c_v x_v + \frac{1}{2} c_v \bar{y}_v$$

$$c_v \bar{x}_v \to \frac{1}{2} c_v \bar{x}_v + \frac{1}{2} c_v y_v$$

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Submodular p/w terms:

$$c_{uv} \bar{x}_u x_v \rightarrow \frac{1}{2} c_{uv} \bar{x}_u x_v + \frac{1}{2} c_{uv} y_u \bar{y}_v$$

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$$\min_{\substack{x,y \in \{0,1\}^{|\mathcal{V}|} \\ y = \bar{x}}} g(x,y) = \min_{x \in \{0,1\}^{|\mathcal{V}|}} E(x)$$

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$$\min_{x,y \in \{0,1\}^{|\mathcal{V}|}} g(x,y) \le \min_{\substack{x,y \in \{0,1\}^{|\mathcal{V}|} \\ y = \bar{x}}} g(x,y) = \min_{x \in \{0,1\}^{|\mathcal{V}|}} E(x)$$

Theorem: $$\min_{x,y \in \{0,1\}^{|\mathcal{V}|}} g(x,y) = \min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle$$

Binary LP Relaxation = O-QPBF Relaxation

Theorem: $$\min_{x,y \in \{0,1\}^{|\mathcal{V}|}} g(x,y) = \min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle$$

$$\mu'_v(1) = \begin{cases} x_v, & x_v = \bar{y}_v \\ \frac{1}{2}, & x_v \neq \bar{y}_v \end{cases},$$

$$\mu'_v(0) = 1 - \mu'_v(1)$$

$$\mu'_{uv} = \arg\min_{\mu_{uv}} \langle \theta_{uv}, \mu_{uv} \rangle$$

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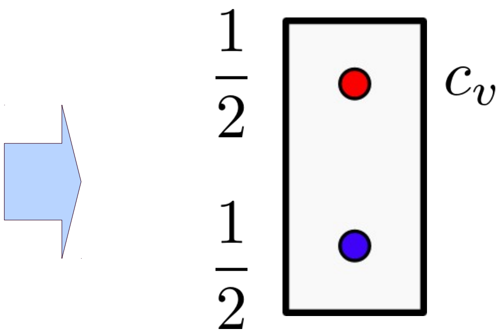
Proof:

1) $x_w = \bar{y}_w$ - holds by construction

2) unary terms:
$x_v \neq \bar{y}_v$

$$c_v x_v \to \frac{1}{2} c_v x_v + \frac{1}{2} c_v \bar{y}_v$$

$$c_v \bar{x}_v \to \frac{1}{2} c_v \bar{x}_v + \frac{1}{2} c_v y_v$$



3) Pairwise terms: $x_v = \bar{y}_v$ and $x_u \neq \bar{y}_u$

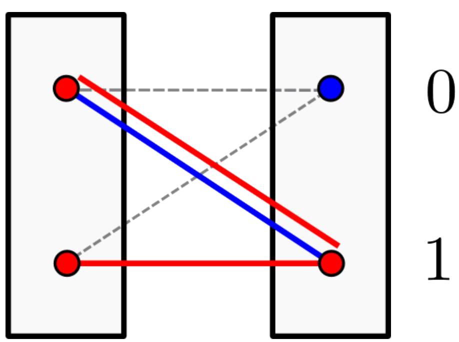
Submodular terms: $c_{uv}\bar{x}_u x_v \to \frac{1}{2}c_{uv}\bar{x}_u x_v + \frac{1}{2}c_{uv}y_u\bar{y}_v$

a) $x_v = \bar{y}_v = 1$ $\qquad \frac{1}{2}c_{uv}(x_u + \bar{y}_u) = \frac{1}{2}c_{uv}$

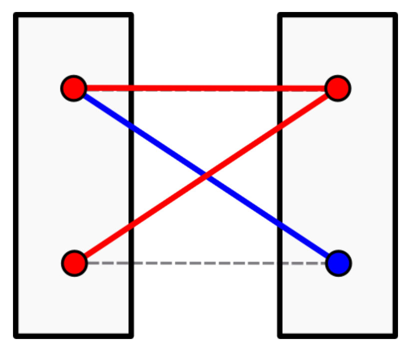
b) $x_v = \bar{y}_v = 0$ $\qquad 0$

$\mu > 0$

$\theta = c_{uv} > 0$



$x_v = \bar{y}_v = 1$



$x_v = \bar{y}_v = 0$

4) Pairwise terms: $x_v = \bar{y}_v \text{ and } x_u \neq \bar{y}_u$

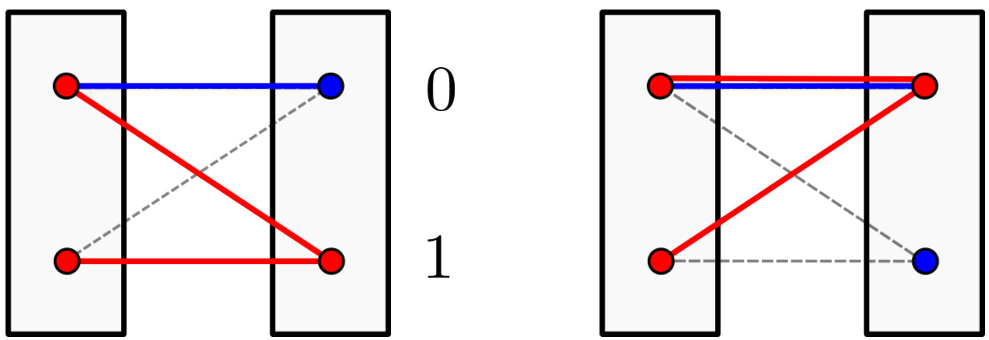
Supermodular terms: $$c_{uv} x_u x_v \to \frac{1}{2} c_{uv} x_u \bar{y}_v + \frac{1}{2} c_{uv} \bar{y}_u x_v$$

a) $x_v = \bar{y}_v = 1$ $\qquad 0$

b) $x_v = \bar{y}_v = 0$ $\qquad \frac{1}{2} c_{uv} (x_u + \bar{y}_u) = \frac{1}{2} c_{uv}$

$\mu > 0$

$\theta = c_{uv} > 0$



$x_v = \bar{y}_v = 1$ $\qquad$ $x_v = \bar{y}_v = 0$

5) Pairwise terms: $x_v \neq \bar{y}_v$ and $x_u \neq \bar{y}_u$

submodular

$$c_{uv}\bar{x}_u x_v \to \frac{1}{2}c_{uv}\bar{x}_u x_v + \frac{1}{2}c_{uv}y_u\bar{y}_v$$

supermodular

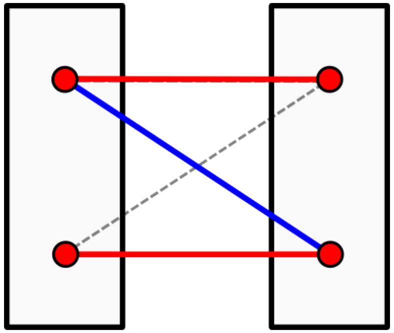
$$c_{uv}x_u x_v \to \frac{1}{2}c_{uv}x_u\bar{y}_v + \frac{1}{2}c_{uv}\bar{y}_u x_v$$

a) $x_u = y_u = x_v = y_v = 0$ $\qquad 0$

b) $x_u \neq \bar{y}_u,\ x_u = y_u = 1$ $\qquad \geq 0$

$\mu > 0$

$\theta = c_{uv} > 0$

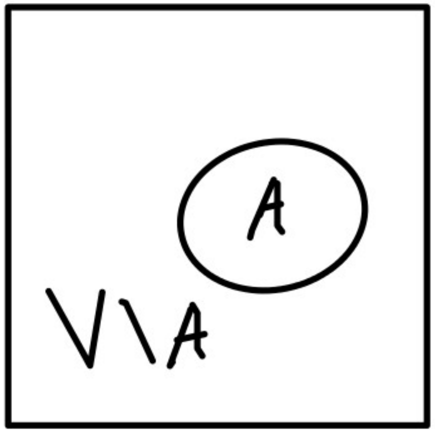


$x_v \neq \bar{y}_v$ and $x_u \neq \bar{y}_u$

b) → **Non-optimal**:
- increases this term

and
- does not influence others

Persistency, Partial Optimality

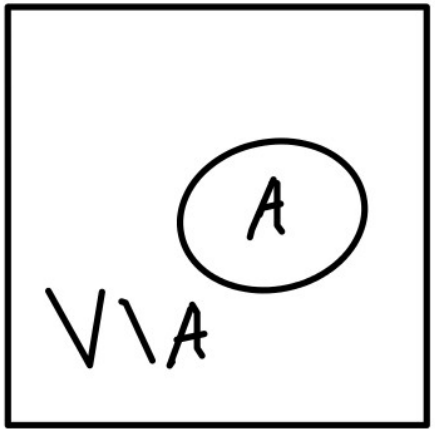


A-integer Labels
V/A-fractional Labels

Definition:

$$x^{\mathcal{A}} - \text{(weakly) persistent, if } x^{\mathcal{A}} = x^*|_{\mathcal{A}}, \text{ where } x^* \in \arg\min_{x \in \mathcal{X}} \langle \theta, \delta(x) \rangle$$

Persistency, Partial Optimality



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Persistency criterion:

$x^{\mathcal{A}}$ – (weakly) persistent, if for all $\tilde{x} \in \mathcal{X}_{\mathcal{V} \setminus \mathcal{A}}$ it holds:

$$x^{\mathcal{A}} \in \arg\min_{x \in \mathcal{X}_{\mathcal{A}}} E((x, \tilde{x}))$$

Theorem: Integer part of any solution to the LP relaxation of binary energy is persistent

$\mathcal{A}(\mu) = \{v \in \mathcal{V} \colon \mu_v \in \{0,1\}^2\}$ - set of integer coordinates

$$\text{Let } \mu' \in \arg\min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle \text{ . Then } \mu'|_{\mathcal{A}(\mu')} \text{ is persistent}$$

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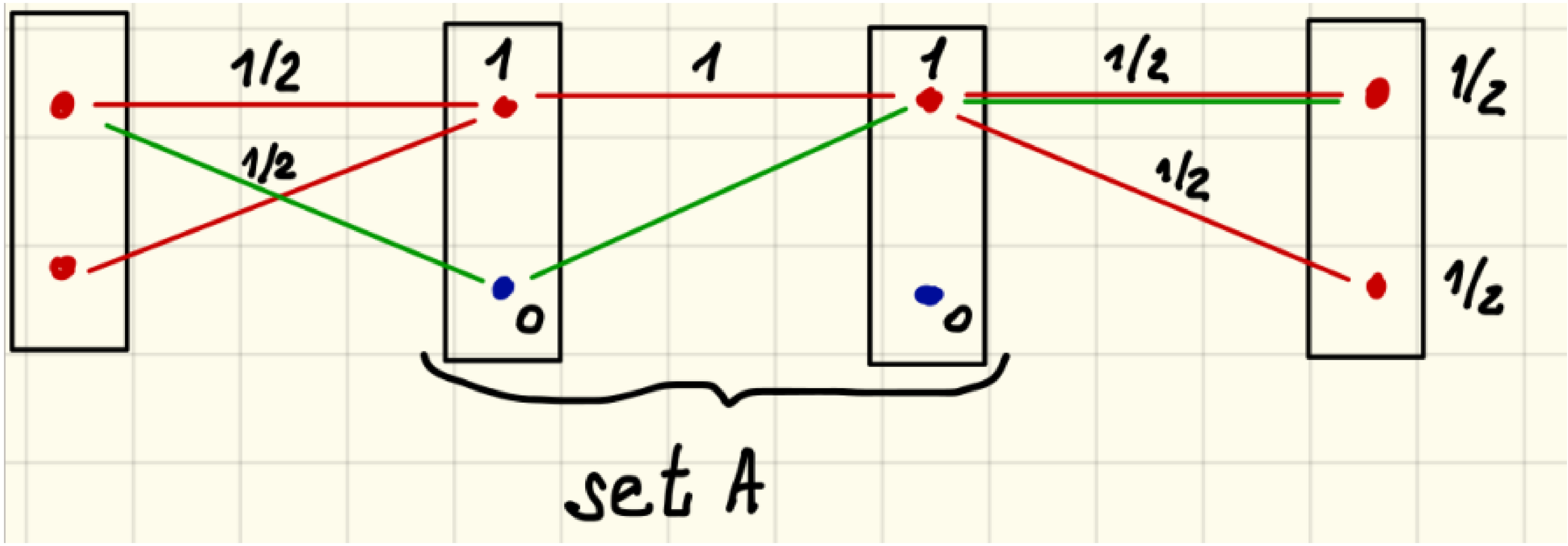
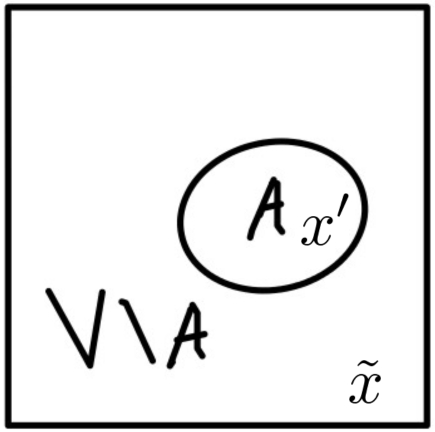


Does the persistency property holds in the multi-label case?

1) yes, there is no difference
2) no, because the proof works only in the binary case
3) no, because we had a counterexample earlier
(partial integrality does not imply partial optimality)
4) no, because I know a counterexample and can give it
5) 2)-3) are both correct
6) 2)-3) are both correct and 4) holds for me

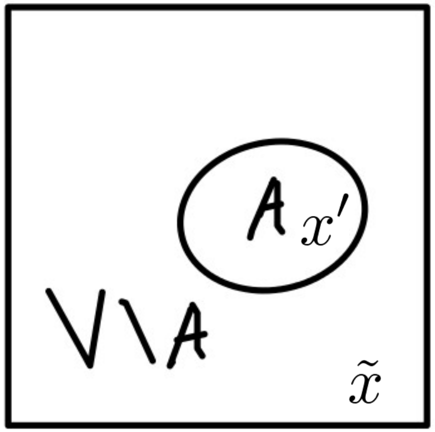


A-integer Labels

V/A-fractional Labels

$$x' = \mu'|_{\mathcal{A}(\mu')}$$

$$E((x, \tilde{x})) \geq E((x', \tilde{x})), \quad \forall x \in \mathcal{X}_{\mathcal{A}} \text{ and } \tilde{x} \in \mathcal{X}_{\mathcal{V} \setminus \mathcal{A}}$$



A-integer Labels

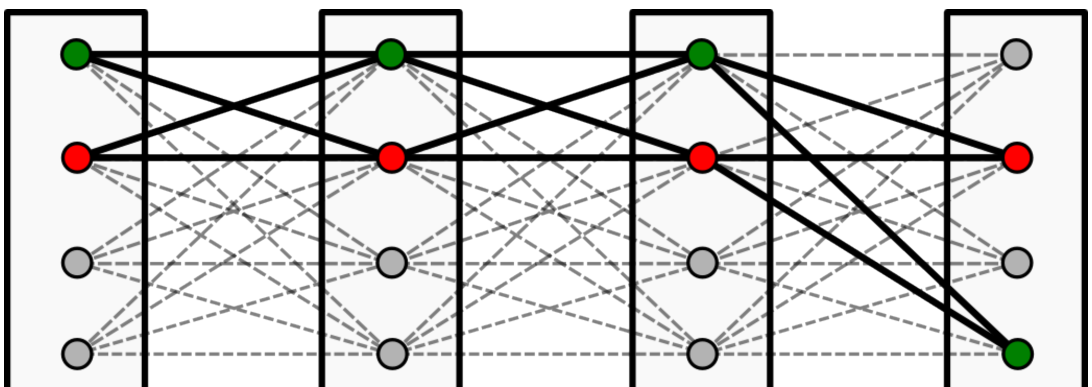
V/A-fractional Labels

$$x' = \mu'|_{\mathcal{A}(\mu')}$$

$$E((x,\tilde{x})) \geq E((x',\tilde{x})), \quad \forall x \in \mathcal{X}_{\mathcal{A}} \text{ and } \tilde{x} \in \mathcal{X}_{\mathcal{V}\setminus\mathcal{A}}$$

α-Fusion moves:

Select α to expand, solve the binary problem



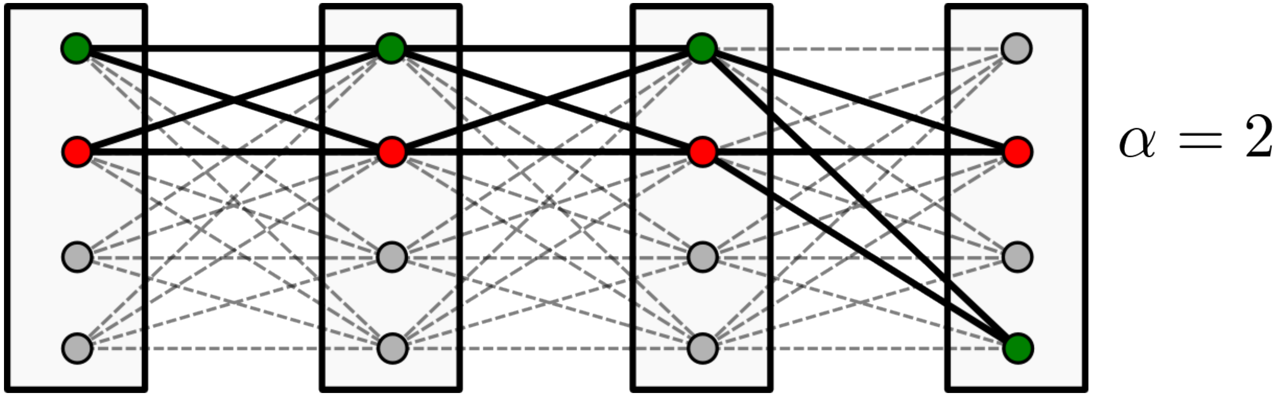
$\alpha = 2$

Properties:

+1) monotonous (energy non-increasing)
+2) applicable to any potentials
+3) other move strategies can be used
- 4) twice as big min-cut subproblems, as in alpha-expansion
- 5) no optimality guarantees
- 6) no optimality bounds

α-Fusion moves:

Select α to expand, solve the binary problem



Conclusions

LP relaxation of binary energies:

1. Arc-consistency = relaxed optimality (can be solved by TRW-S)
2. Can be solved by min-cut
3. Has a persistency property
4. Can be used for monotonous fusion move algorithms